



Power Functions

Graphs of power functions differ in shape, depending on the degree of the polynomial.

You can predict how the graph of a function will look and behave if you know its degree and leading coefficient.

Identifying Graphs of Power Functions

Definition

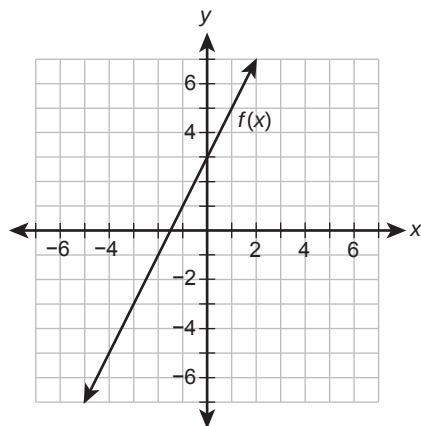
A **power function** is any function that can be written in the form

$$f(x) = ax^n + b$$

where n is a positive integer, a is any nonzero real number, and b is any real number.

Graphs of linear, quadratic, and cubic functions differ greatly in their appearance.

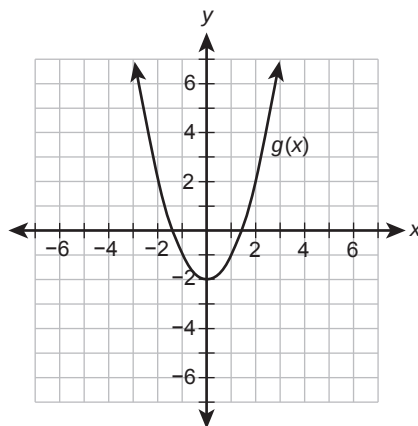
**Linear
Degree 1**



$$f(x) = 2x + 3$$

The graph of a linear function is a line.

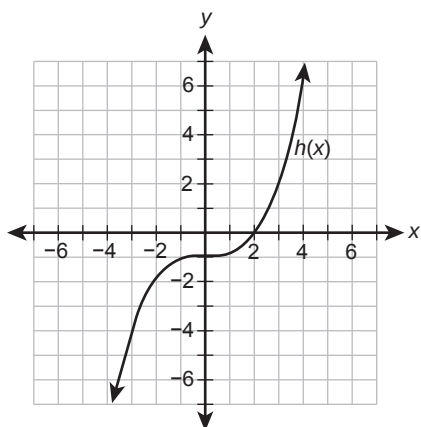
**Quadratic
Degree 2**



$$g(x) = x^2 - 2$$

The graph of a quadratic function is a parabola.

**Cubic
Degree 3**



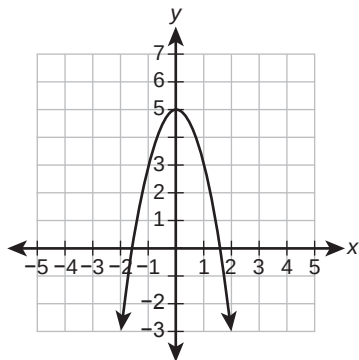
$$h(x) = \frac{1}{9}x^3 - 1$$

The graph of a cubic function is shaped like a sideways S.

EXAMPLE 1

Determine whether the graph is a linear, quadratic, or cubic function.

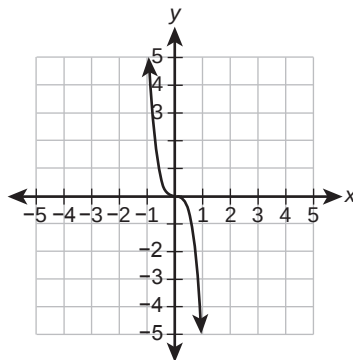
A



SOLUTION

The function is a parabola, so it is quadratic.

B



SOLUTION

The function is S-shaped, so it is cubic. ■

Graphing Power Functions

EXAMPLE 2

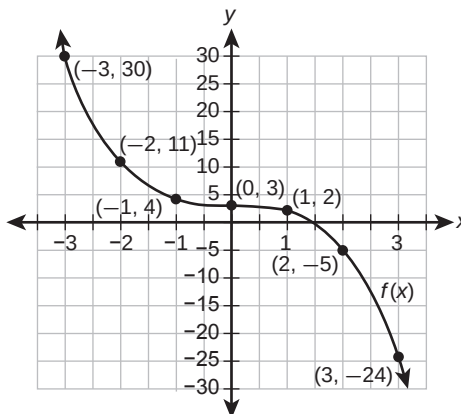
A Graph $f(x) = -x^3 + 3$.

SOLUTION

Step 1 Make a table of values.

x	-3	-2	-1	0	1	2	3
$f(x)$	30	11	4	3	2	-5	-24

Step 2 Graph the function. Plot each point and join the points with a smooth curve.



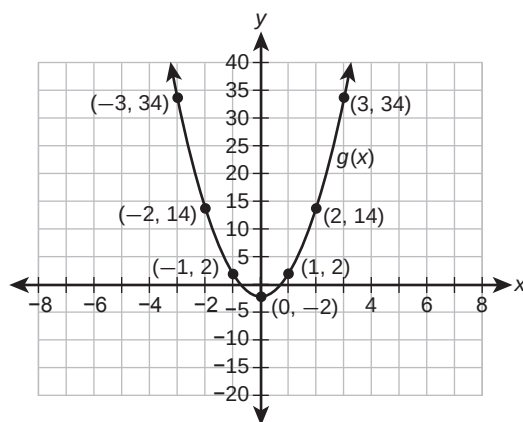
B Graph $g(x) = 4x^2 - 2$.

SOLUTION

Step 1 Make a table of values.

x	-3	-2	-1	0	1	2	3
$g(x)$	34	14	2	-2	2	14	34

Step 2 Graph the function. Plot each point and join the points with a smooth curve.



Describing the End Behavior of a Polynomial Function

Knowing what to expect for the end behavior of a graph can help you visualize a polynomial function.

Definition

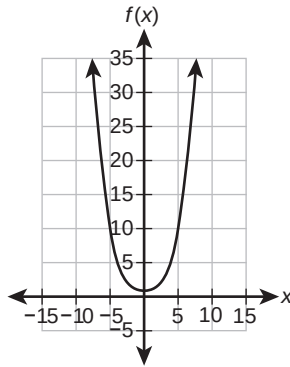
The **end behavior** of a function is how the function behaves when the domain values increase or decrease without bound.

The end behavior of a polynomial function depends on the degree of the function and on the sign of the leading coefficient.

End Behavior of Power Functions

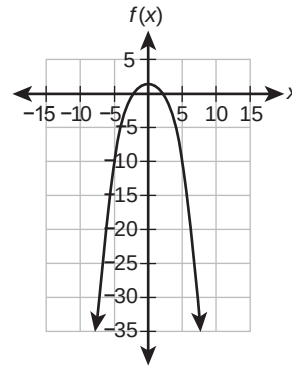
End Behavior of Power Functions: $f(x) = ax^n + b$

n is even, $a > 0$



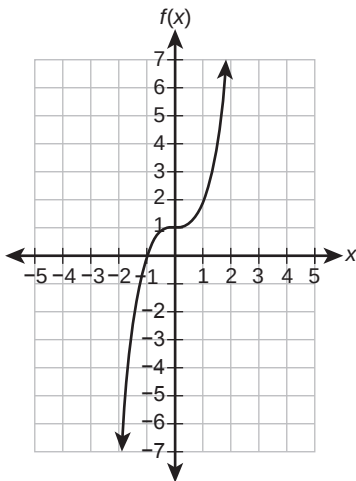
As x increases without bound, $f(x)$ eventually increases without bound. As x decreases without bound, $f(x)$ eventually increases without bound.

n is even, $a < 0$



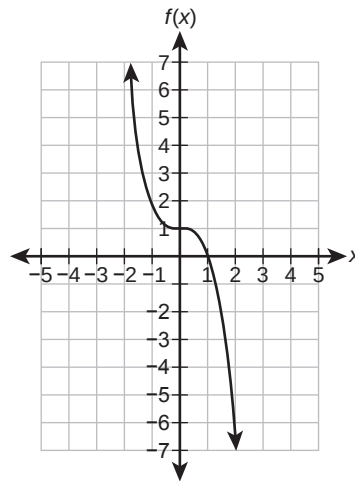
As x increases without bound, $f(x)$ eventually decreases without bound. As x decreases without bound, $f(x)$ eventually decreases without bound.

n is odd, $a > 0$



As x increases without bound, $f(x)$ eventually increases without bound. As x decreases without bound, $f(x)$ eventually decreases without bound.

n is odd, $a < 0$



As x increases without bound, $f(x)$ eventually decreases without bound. As x decreases without bound, $f(x)$ eventually increases without bound.

EXAMPLE 3

Describe the end behavior of the function.

A $f(x) = -2x^4 + 7$

SOLUTION

The function has degree 4 and the leading coefficient is negative, so as x increases without bound, $f(x)$ eventually decreases without bound, and as x decreases without bound, $f(x)$ eventually decreases without bound.

B $f(x) = x^3 - 1$

SOLUTION

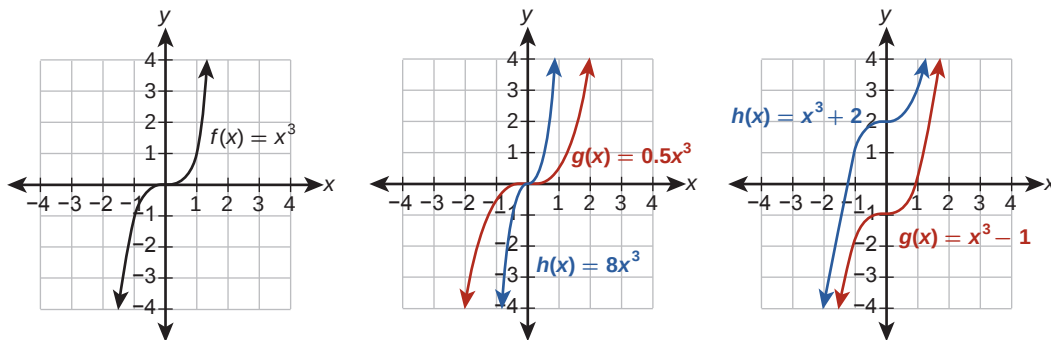
The function is cubic and the leading coefficient is positive, so as x increases without bound, $f(x)$ eventually increases without bound, and as x decreases without bound, $f(x)$ eventually decreases without bound. ■

Relating Power Functions Within a Family of Graphs

Power Functions

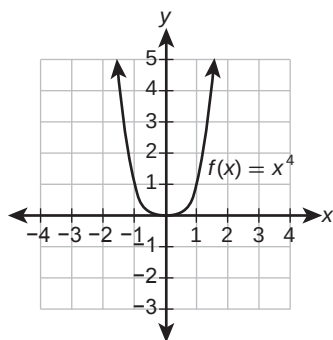
Power Function Graph Family: $f(x) = ax^n + b$

Each change in the value of a or b results in a different transformation of the parent graph, $f(x) = x^n$.

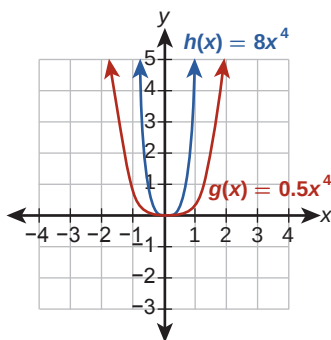


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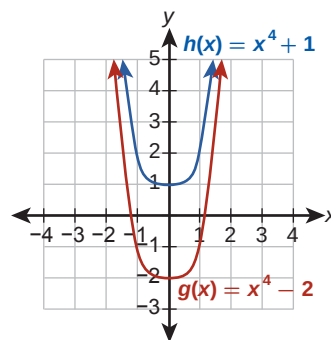
Power Functions (Continued)



All equations of the form $f(x) = x^n$ go through the origin.



If you change only the parameter a , the graphs expand or contract while passing through the origin.



If you change only the parameter b , the graphs shift up or down on the y -axis, resulting in different y -intercepts.

Writing a Power Function When Given a Point

EXAMPLE 4

Find the equation for the power function of the form $p(x) = ax^2$ that contains the point $(-2, 12)$.

SOLUTION

$$p(x) = ax^2$$

$$12 = a \cdot (-2)^2 \quad \text{Substitute 12 for } p(x) \text{ and } -2 \text{ for } x.$$

$$\frac{12}{4} = a \quad \text{Solve for } a.$$

$$3 = a$$

Substitute the value found for a into $p(x) = ax^2$. The function is $p(x) = 3x^2$. ■

Identifying Odd and Even Functions

You can also determine whether a function is even or odd.

Definitions

A function g is even if $g(-x) = g(x)$ for all values of x . The graph of an even function is symmetric about the y -axis.

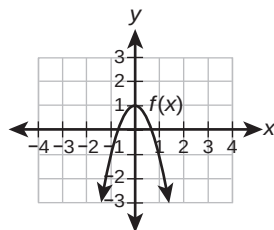
A function h is odd if $h(-x) = -h(x)$ for all values of x . The graph of an odd function is symmetric about the origin.

EXAMPLE 5

Identify the function as even, odd, or neither.

A $f(x) = -2x^2 + 1$

SOLUTION
geometrically



The function is symmetric about the y -axis.

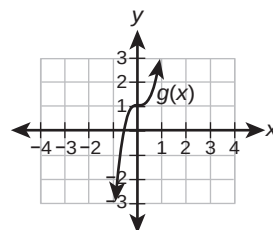
algebraically

$$\begin{aligned}f(-x) &= -2(-x)^2 + 1 \\&= -2x^2 + 1 \\&= f(x)\end{aligned}$$

The function $f(x) = -2x^2 + 1$ is an even function.

B $g(x) = 7x^3 + 2$

SOLUTION
geometrically



The function is not symmetric about the y -axis or the origin.

algebraically

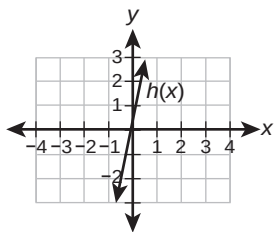
$$\begin{aligned}g(-x) &= 7(-x)^3 + 2 \\&= -7x^3 + 2\end{aligned}$$

The function $g(x) = 7x^3 + 2$ is neither even nor odd.

C $h(x) = 5x$

SOLUTION

geometrically



The function is symmetric about the origin.

algebraically

$$\begin{aligned} h(-x) &= 5(-x) \\ &= -5x \\ &= -h(x) \end{aligned}$$

The function $h(x) = 5x$ is an odd function. ■

Using a Table of Values to Classify a Polynomial

By finding the successive differences when given a table of values, you can determine what degree the power function has. First differences are the differences of the function values. Second differences are the differences of the first differences. Third differences are the differences of the second differences, and so on.

linear $f(x) = 3x + 2 = 3x^1 + 2$ has **1st** successive differences equal.

quadratic $f(x) = x^2 - 4x - 5$ has **2nd** successive differences equal.

cubic $f(x) = 2x^3 + 9$ has **3rd** successive differences equal.

quartic $f(x) = 5x^4 - 3x^2 + 1$ has **4th** successive differences equal.

Successive Differences of Power Functions

An n th-degree power function has a constant n th successive difference.

EXAMPLE 6

Determine whether the table of values represents a linear, quadratic, or cubic function.

A

x	-2	-1	0	1	2
$f(x)$	-8	-6	-4	-2	0

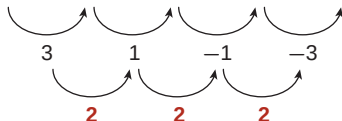


SOLUTION

The differences of successive $f(x)$ values are equal, so this table of values represents a linear function.

B

x	-2	-1	0	1	2
$g(x)$	7	4	3	4	7

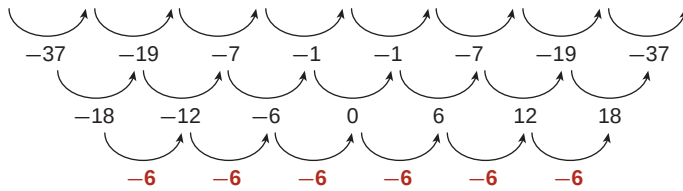


SOLUTION

The successive differences of $g(x)$ values are not equal, so find the differences of the first differences. Because these second differences are equal, the table of values represents a quadratic function.

C

x	-4	-3	-2	-1	0	1	2	3	4
$h(x)$	-64	-27	-8	-1	0	1	8	27	64

**SOLUTION**

The successive differences of $h(x)$ are not equal and the second differences are not equal, so find the third differences. The third differences are equal, so the table of values represents a cubic function. ■